CS & IT ENGINEERING



Algorithms

Dynamic Programming
Discussion Notes

DPP – 01



By- Aditya sir

[MCQ]

#Q. What is the time complexity of dynamic programming for matrix chain multiplication problem?

A $O(n^2)$

B $O(n^3)$

C $O(n\log n)$

D None of these

Ans : B

Matrix Chain Multiplication

MCM.

$(A_1 A_2 \ldots A_n)$

n → no. of matrices

$\Rightarrow O(n * n^2)$

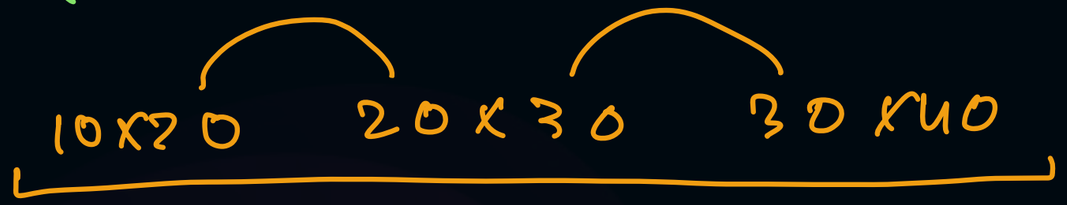
$\Rightarrow \boxed{O(n^3)}$

DP based mcm → $O(n^3)$ ✓

Space Complexity → $O(n^2)$ ✓

[NAT]

#Q. Consider the matrices x, y and z with dimension 10 × 20, 20 × 30 and 30 × 40 respectively. Then what is the minimum number of multiplications required to multiply the matrices? _____



Matrix Chain Multiplication (MCM)

(x y z)

Ans :- 18000

1

((x y) z)

18000

2

(x (y z))

32000

① $(XY)Z$

$\Rightarrow$

(XY) → 10×20, 20×30

10×20 $\quad$ 20×30

10×30 $\quad$ 30×40 = $10\times20\times30 + 10\times30\times40$

$X \rightarrow 10\times20$

$Y \rightarrow 20\times30$

$Z \rightarrow 30\times40$

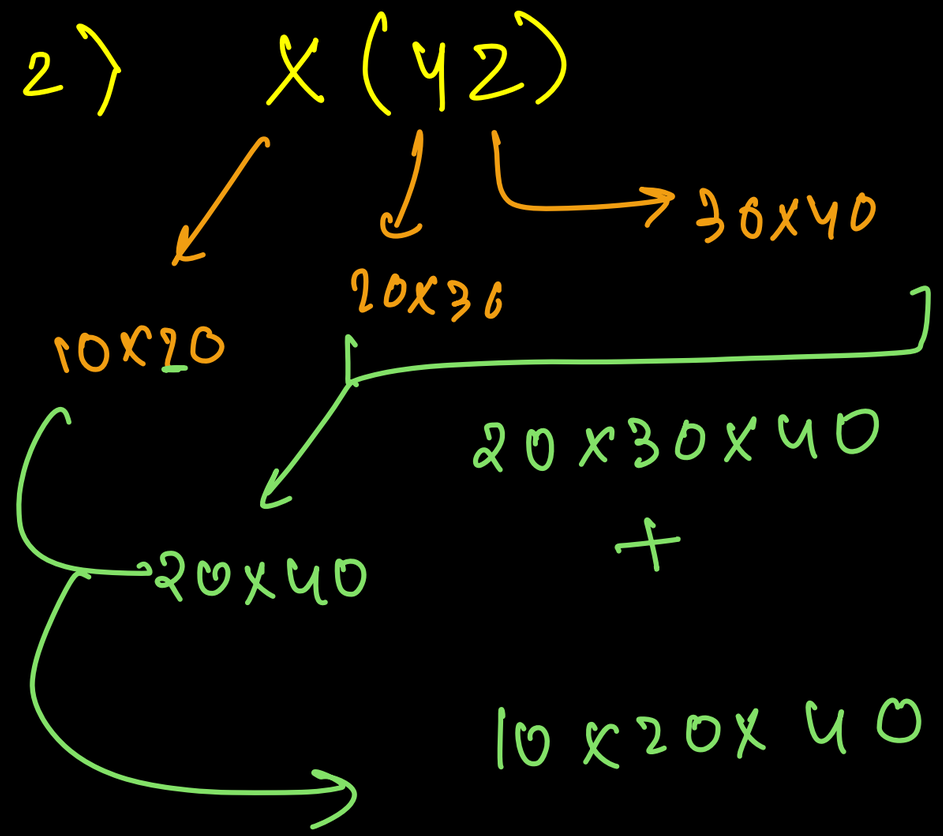
$A_{m\times n} * B_{n\times p} = C_{m\times p}$

no. of mul = $m\times n\times p$

$= 6000 + 12000$

$= 18000$

2) X(42)



= 24000
+
8000

= 32000

[MCQ]

#Q. What is the length of the LCS for the pair of strings given below.

P = ATGACTATAA

Q = GACTAATA

A. 5 ✗

B. 6 ✗

C. 7 ✓

D. 8 ✗

Longest Common Subsequence

(LCS)

Ans: C

Approaches:

1) manually

2) Top-Down

3) Bottom-up



Idea :- $A[1 \ldots n]$ $B[1 \ldots m]$

$i \rightarrow$ $\quad$ $j \rightarrow$

$$LCS(i,j) = 1 + LCS(i-1, j-1), \quad A[i] = B[j]$$

$$LCS(i,j) = \max \begin{Bmatrix} LCS(i-1, j) \\ LCS(i, j-1) \end{Bmatrix}, \quad A[i] \neq B[j]$$

Soln : Bottom - up = Tabulation.

P = ATGACTATAA

Q = GACTAATA

			A	T	G	A	C	T	A	T	A	A
		0	1	2	3	4	5	6	7	8	9	10
	0	0	0	0	0	0	0	0	0	0	0	0
G	1	0	0	0	1	1	1	1	1	1	1	1
A	2	0	1	1	1	2	2	2	2	2	2	2
C	3	0	1	1	1	2	3	3	3	3	3	3
T	4	0	1	2	2	2	3	4	4	4	4	4
A	5	0	1	2	2	3	3	4	5	5	5	5
A	6	0	1	2	2	3	3	4	5	5	6	6
T	7	0	1	2	3	3	3	4	5	6	6	6
A	8	0	1	2	3	4	4	4	5	6	7	7

Ans

LCS : GACTATA $\Rightarrow$ (7)



match : $1 + L[i-1, j-1]$ (diagonal)

not match : $\max\{\text{Left}, \text{Up}\}$

Appr2 : Manually:

P = A T G A C T A T A A ⟶ n = 10

Q = G A C T A A T A ⟶ m = 8

1 2 3 - - . . 10 X

min (m, n) = min (10, 8) = 8 X

X

8 7 6 5 4 3 2 1 0

G A C T A T A C

[MCQ]

#Q. Consider a connected weighted graph G = (V, E),where |V| = n, |E| = m, if all the edges have distinct positive integer weights, then the maximum number of minimum weight spanning trees in the graph is ?

A. n

B. m

C. 1

D. n^{n-2}

Ans: C

n → vertices

m → edges.

Distinct positive wts

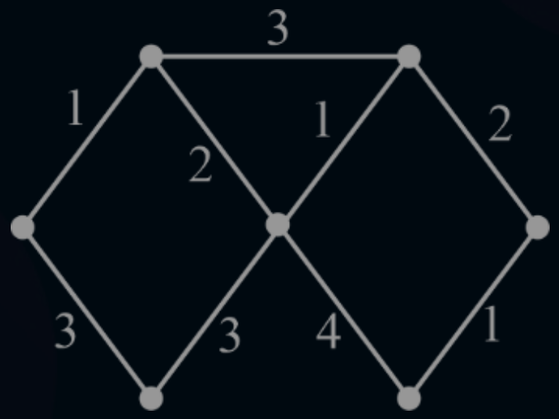
↳ Prims
↳ Kruskal

} Structure & Min Cost

↳ both same.

[NAT]

#Q. What is the weight of the minimum spanning tree for the graph shown below?



Ans: 10

Boruvka

Dijkstra MCST

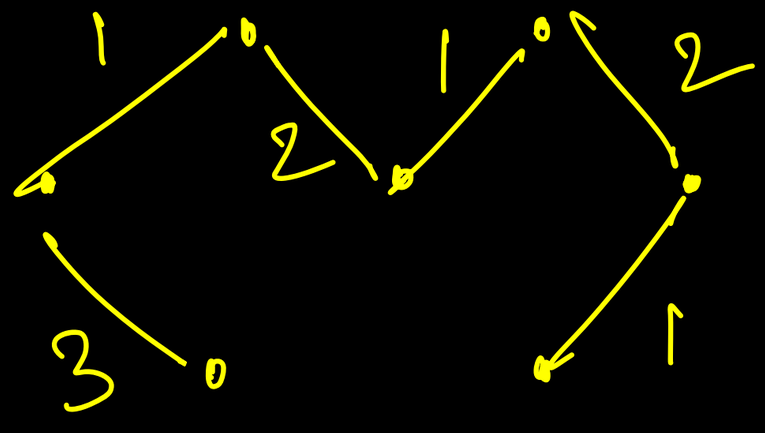


MCST cost = 1+1+1+2+2+3

= 10

Soln :-

Kruskal :

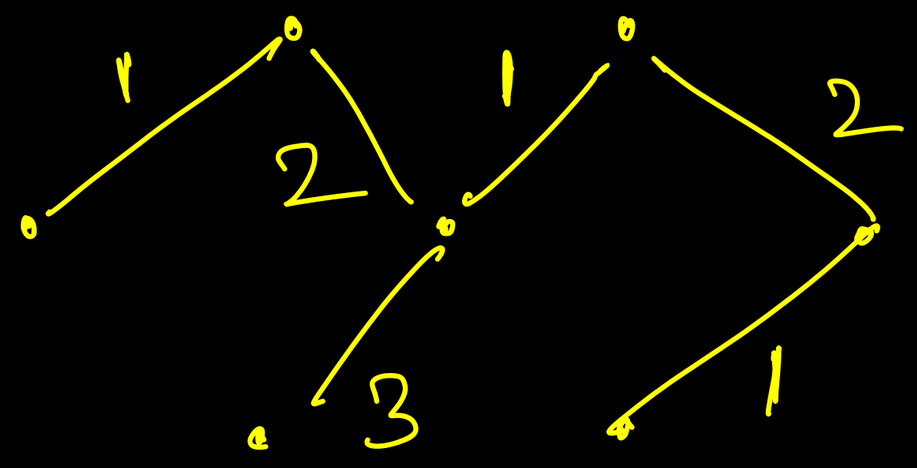


MCST Cost = 1+1+1+2+2+3

= 3+4+3

= 10

Prims



MCST cost = 1+1+1+2+2+3

= 10

[MCQ]

#Q. How many minimum spanning tree does this graph have?

A 2

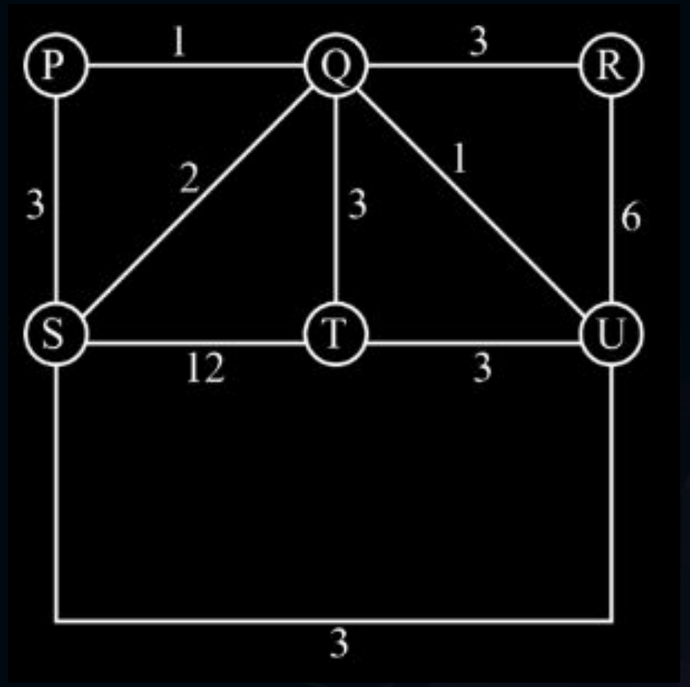
B 3 ✗

C 4 ✗

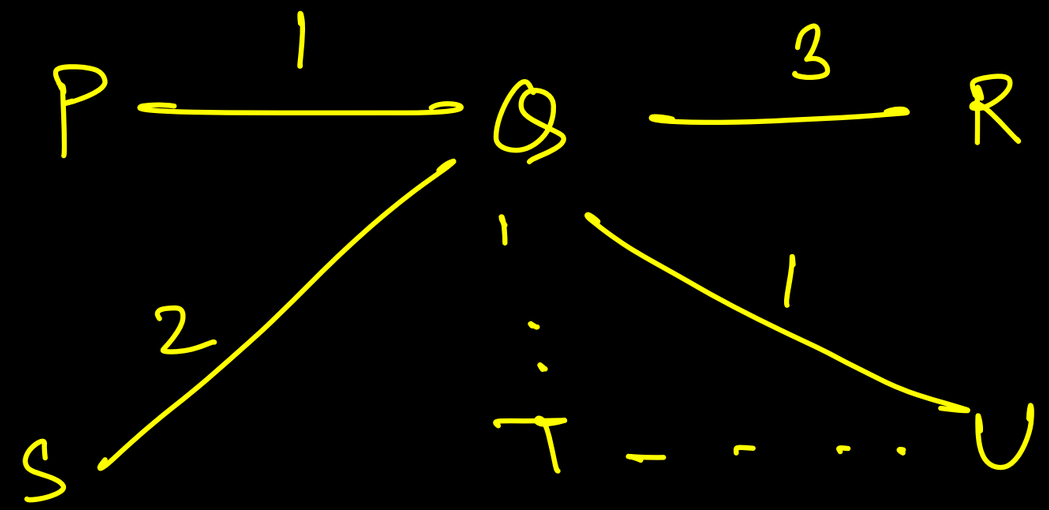
D 5 ✗

2 MCSTs

Ans: A



Soln :-

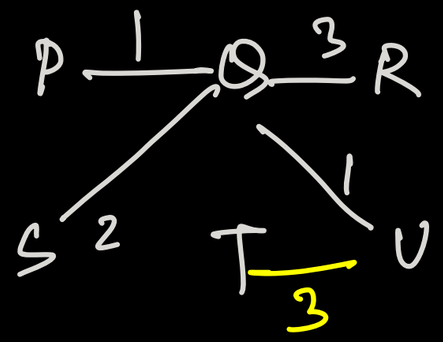


$n = 6$

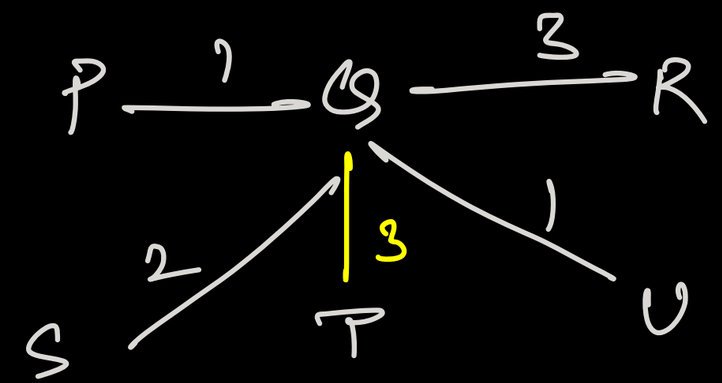
edges in MCST $= 6 - 1$

$= \boxed{5}$

${}^2C_1$



OR



2 MCSTs

[MCQ]

#Q. Consider the following problem with knapsack capacity of 8

Item	Profits	Weights
I_1	13	1
I_2	8	5
I_3	7	3
I_4	3	4

$n = 4$
$M = 8$

Which of the following item is not selected in the optimal solution of 0/1, knapsack problem?

C I_1 only

C I_2 only

D I_3 only

D I_4 only

Ans: C

0/1 → Binary knapsack ⟶ Tabulation.

n ⟶ (n+1) x (m+1)
m

Recurrence :-

$$Knap(i,j) = Knap(i-1,j) \quad , \; w[i] > M$$

$$= \max \left\{ \begin{array}{l} Knap(i-1,j) \\ Knap(i-1, j-w[i]) + P[i] \end{array} \right\}$$

when $w[i] \leq M$

B → m
v → n

m

Obj n

	P	W	
			0
1	13	1 ✓	1 →
2	8	5	2
3	7	3 ✓	3
4	3	4 ✓	4

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	
1	0	13	13	13	13	13	13	13	13	
2	0	13	13	13	13	13	21	21	21	
3	0	13	13	13	20	20	21	21	21	
4	0	13	13	13	20	20	21	21	23	

max

Ans :-

I_1, I_3, I_4 → optimal Profit

[MCQ]

#Q. Consider the following statements

S1: for every weighted graph and any two vertices p and q, Bellman ford algorithm starting at p will always return a shortest path to q.

S2: Dijkstra greedy algorithm for single source shortest path can be used to solve the all pairs shortest path problem.

Which of the statement is correct?

C only S_1

C only S_2

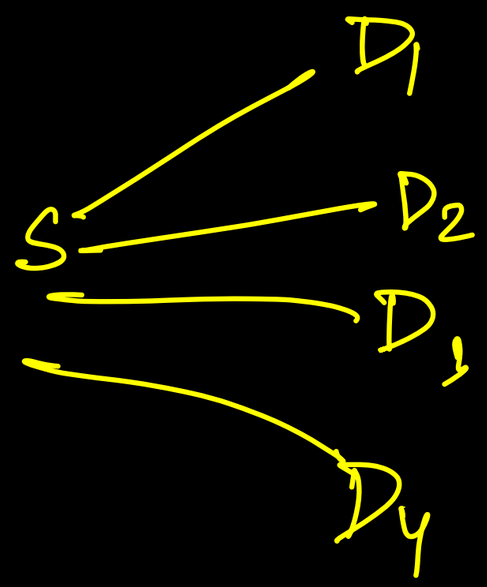
D Both S_1 and S_2 are true

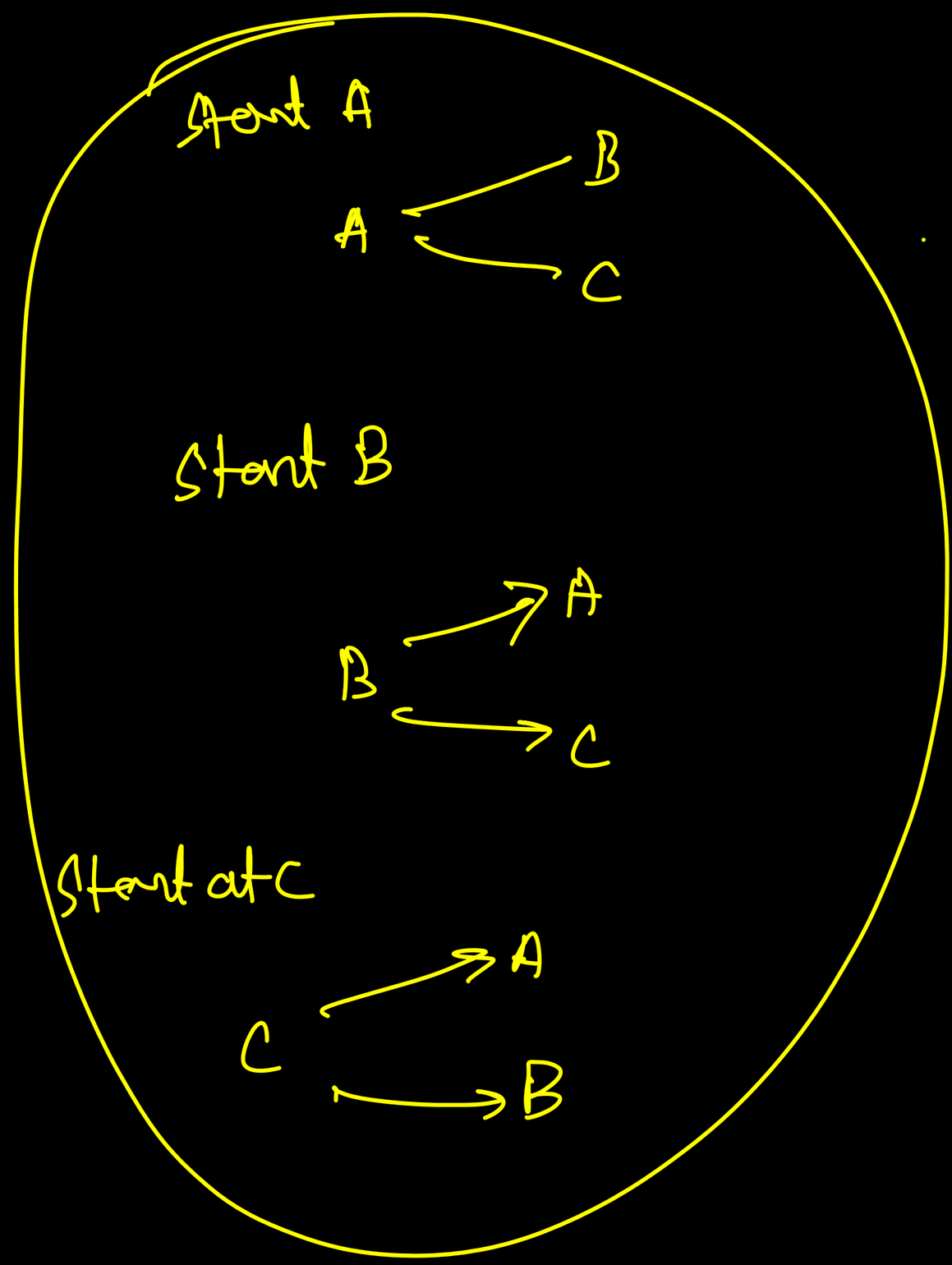
D neither S_1 nor S_2 is true

Single Source Shortest Path :-

		not +ve edges	-ve not edges	-ve not cycle
1)	Dijkstra → Greedy	✓	✗ But no -ve edges	✗
2)	Bellman Ford → DP	✓	✓	✗

SSSP D1





APSP

THANK - YOU